

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

## 4<sup>th</sup> Semester of B. Pharm. Examination University Theory Examination May 2017 PH223 Pharmaceutical Biochemistry

Date: 15.05.2017, Monday

Time: 1.30 pm to 4.30 pm

Maximum Marks: 80

### Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

### SECTION - I

**Q 1** Attempt all questions. Each question is of one mark. **20**

1. \_\_\_\_\_ inhibitors are used to prevent retinopathy in diabetic patient.  
[A] aldose reductase  
[B] ketosereductase  
[C] sorbito dehydrogenase  
[D] glucose-6-phosphate dehydrogenase
2. Which of the following is Monoenoic acids ?  
[A] Palmitoleic acid  
[B] Oleic Acid  
[C] Elaidic acid  
[D] All of above
3. In sugar phosphate backbone, sugars are linked together by  
[A] disulfide bond  
[B] sulfite bond  
[C] ester bond  
[D] phosphodiester bond
4. the primary periodicity in X ray differection pattern of DNA is  
[A] 3.4 Å  
[B] 34 Å  
[C] 15 Å  
[D] 10 Å
5. Which of the following enzyme is not involved in HMP shunt?  
[A] Glyceraldehyde-3-p dehydrogenase  
[B] Glucose-6-p-dehydrogenase  
[C] Transketolase  
[D] Phosphogluconate dehydrogenase
6. which of the following amino acid can act as neucleophilic catalyst during enzymatic reaction?  
[A] phenylalanine  
[B] serine  
[C] histidine  
[D] proline
7. \_\_\_\_\_ is an optically inactive amino acid is  
[A] serine  
[B] glycine  
[C] threonine  
[D] valine

8. Coenzyme Q is involved in electron transport as
  - [A] carrier of O<sub>2</sub>
  - [B] a water-soluble electron donor
  - [C] covalently attached cytochrome cofactor
  - [D] a lipid-soluble electron carrier
9. A derivative of pyridoxine (vitamin B<sub>6</sub>) is required as a coenzyme in:
  - [A] transamination
  - [B] oxidative deamination
  - [C] the urea cycle
  - [D] carbamoyl phosphate formation
10. Which one is the heaviest particulate component of the cell?
  - [A] Nucleus
  - [B] Mitochondria
  - [C] Cytoplasm
  - [D] Golgi apparatus
11. Which one of the following is a rate limiting enzyme of gluconeogenesis?
  - [A] Hexokinase
  - [B] Phosphofructokinase
  - [C] Pyruvate carboxylase
  - [D] Pyruvate kinase
12. Acetyl CoA is not used for the synthesis of
  - [A] Fatty acid
  - [B] Cholesterol
  - [C] Pyruvic acid
  - [D] Citric acid
13. An allosteric enzyme responsible for controlling the rate of T.C.A cycle is
  - [A] Malate dehydrogenase
  - [B] Isocitrate dehydrogenase
  - [C] Fumarase
  - [D] Aconitase
14. The chief protein of cow's milk is
  - [A] Albumin
  - [B] Vitellin
  - [C] Livetin
  - [D] Casein
15. From dietary protein as well as from the urea present in fluids secreted into the gastrointestinal tract intestinal bacteria produce
  - [A] Carbon dioxide
  - [B] Ammonia
  - [C] Nitrogen
  - [D] Creatine
16. One mol. of Urea is synthesized at the expense of the \_\_\_\_\_ mols. of ATP.
  - [A] 2
  - [B] 3
  - [C] 4
  - [D] 5
17. Neutral amino acid is
  - [A] Leucine
  - [B] Lysine
  - [C] Aspartic acid
  - [D] Histidine
18. Number of carbon atoms in cholesterol is
  - [A] 17
  - [B] 19
  - [C] 27
  - [D] 30
19. Ketone bodies are synthesized from fatty acid oxidation products by which of the following organs?
  - [A] Liver
  - [B] Skeletal muscles
  - [C] Kidney
  - [D] Brain
20. LCAT is \_\_\_\_\_
  - [A] Lactose choline alaminotransferase
  - [B] Lecithin cholesterol acyl transferase

## SECTION – II

- Q 2** Attempt any **TWO** of the following
- A** Describe metabolic pathway for purine nucleotides. **05**
  - B** Explain various types of bonding interactions in nucleic acid structure. **05**
  - C** Write down various applications of enzymes. **05**
- Q 3** Attempt any **TWO** of the following
- A** Classify amino acids on the basis of structure, nutritional requirements and metabolic fate. **05**
  - B** Draw a labeled diagram of prokaryotic and eukaryotic cell and differentiate them. **05**
  - C** Classify various metabolic disorders associated with carbohydrates. **05**

## SECTION – III

- Q 4** Attempt any **FOUR** of the following
- A** Explain urea cycle with diagram. **05**
  - B** What are carbohydrates? Classify them in detail. **05**
  - C** Explain  $\beta$  oxidation of lipid. **05**
  - D** What are ketone bodies? How they are generated? Explain their significance? **05**
  - E** List out various Factors affecting enzymetic action and explain any two in detail. **05**
  - F** Write a note on oxidative phosphorylation. **05**
- Q 5** Attempt any **FOUR** of the following
- A** Explain different structures of protein with examples. **05**
  - B** Write a detailed note on glycolysis. **05**
  - C** Write the pathway for biosynthesis of cholesterol. **05**
  - D** Draw a pathway for denovo synthesis of purine nucleotides. **05**
  - E** Differentiate enzyme and coenzyme. **05**
  - F** Enzyme involved in respiratory chain control. **05**
-